

Almost every gene set reads higher in the index arm: a size-matched empirical null for small rare-tumour expression series, and what it leaves of the EWSR1::NR4A3 direct-target catalogue

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Running title: A size-matched null for rare-tumour gene-set reads

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Abstract

On a small rare-tumour expression series, almost every gene set anyone scores comes back higher in the index arm. In a 10-versus-6 extraskeletal myxoid chondrosarcoma (EMC) series, PPAR γ targets, hypoxia metagenes and adipogenesis all move alike. The reason is size, not biology: a set's per-sample score is one draw from a distribution whose width depends on the set's **size** and on the platform, so an arbitrary set can print $t = 3.16$ and be indistinguishable from a random one of the same size. **We supply the calibration that refuses such a read** — a size-matched empirical null drawn from the platform's own genes — and apply it to this disease's best-warranted set. Across 2,276 retrieved documents, **the set with a DNA-binding assay against an NR4A3 chimera is three genes: *SEMA3C*, *PPARG*, *ENO3***. Three is below the four-gene floor for a set score, so the set scored is those three pooled with the sixteen native-NR4A3 targets. On the two readable array platforms **that pooled set reaches 39% and 88% of its null threshold and does not clear**, while the published EMC phenotype clears it 11.9-fold and 4.2-fold in the same run — the instrument reads this disease, not this set. Exact label permutation, every comparator stratum separately, a matrix covariate and a muscle control then separate the three: *SEMA3C* survives nothing and reverses sign with the comparator; *PPARG*'s strongest reading is circular, scored on the cohort that first published it; *ENO3* survives everything, but was the pre-designated positive control and is not an independent finding here. **The binding constraint is not sample size**. No experiment has measured where an NR4A3 fusion binds, or what chromatin does, in EMC material — the one genome-wide chromatin readout that exists for these fusions reads *accessibility* in HEK293T (GSE243553), and the 110 NR4A peak sets that exist are the wrong protein or the wrong disease, with no class-A gene carrying unusual occupancy against a background panel. Until a fusion cistrome in EMC chromatin exists, "elevated in EMC" and "driven by the fusion" are inseparable. The null is not EMC-specific: any series with a small index arm and heterogeneous comparators fails the same way.

1 • Introduction

1.1 • Almost every gene set reads higher in the index arm

A rare tumour affords a small index arm and a heterogeneous comparator arm, and on such a series almost every gene set anyone scores comes back higher in the index arm. In the series this paper works with — 10 extraskeletal myxoid chondrosarcomas (EMC) against 6 comparators, GSE4303 on platform GPL3290 — PPAR γ targets, hypoxia

metagenes, adipogenesis, chondroitin-sulfate biosynthesis and arginine metabolism all come back "higher in EMC". Sets with no biological relationship to one another move the same way and by similar amounts.

The reason is not biology. A raw Welch contrast on the sample means uses the samples as its unit of variability and ignores that a set's per-sample score is one draw from a distribution whose width depends on the set's size and on the platform. At $n = 10$ versus 6 that width is large: the 95% band for an arbitrary 17-gene set on GPL3290 is $[-0.297, +0.376]$ SD, so a set can print $t = 3.16$ and remain indistinguishable from a random set of the same size. Seventeen is the size scored there: the pooled direct-target set names nineteen genes, and that platform reads two of them, *ICAM1* and *MYH7*, on no probe.

No read on such a series is interpretable until it is calibrated against a size-matched random gene set drawn from the same platform's own genes. That calibration is the instrument this paper supplies; it costs one seeded resampling, it is drawn in **Figure 1**, and it is applied to this work's own headline result rather than only to other people's. **It is not specific to this disease or this gene set** — any series with a small index arm and a heterogeneous comparator arm has the same failure mode, and rare tumours are where such series are the only ones that exist.

The rest of the paper is that instrument applied to one worked example, chosen because it is the gene set in this disease carrying the strongest mechanistic warrant, and because its literature is small enough to enumerate exhaustively rather than sample. **The example is not the contribution**, and §4.2 states which parts of it would and would not survive being wrong.

1.2 · The disease and the driver

EMC is a rare soft-tissue sarcoma defined by rearrangement of *NR4A3* (NOR-1/TEC). Subramanian *et al.* describe it as "characterized by a balanced translocation most commonly involving $t(9;22)(q22;q12)$ " (PMID 15920699), which produces *EWSR1::NR4A3*; Brenca *et al.* express and assay both that chimera and *TAF15::NR4A3*, "the commonest *TAF15* (exons 1–6)–*NR4A3* (exons 3–8) fusion" (PMID 31020999), with a rarer $t(3;9)(q11-12;q22)$ *TFG::NR4A3* variant accounting for part of the remainder. *NR4A3* is an orphan nuclear receptor, and the chimera places its DNA-binding domain under a strong FET-family transactivation domain. The disease's central molecular hypothesis is therefore straightforward: the fusion is a transcription factor with an aberrant output, and that output is where the disease lives.

1.3 · The gap in the worked example

The hypothesis dates from the fusion's cloning in 1995 (PMID 8634690), and the evidence under it is thin in a specific, checkable way. Two questions bear on it, and they are different questions:

1. Which genes has anyone shown an *NR4A3* chimera to physically bind and drive?
2. Which genes are high in EMC tumours?

Neither is much discussed. Measured against Europe PMC on 2026-08-08: of 1,305 records naming the disease, 261 are reviews, and the three genes with a published fusion DNA-binding assay are named in **3, 1 and 0 of those 261 reviews** respectively (*PPARG*, *SEMA3C*, *ENO3*; 37, 6 and 2 records in the corpus as a whole, and both *ENO3* records are about a different disease). The primary sources are, by contrast, ordinary references of this literature — 42–70% of their citations come from EMC records, and three of the four are cited by four to six EMC reviews. The fourth, the *ENO3* source, is cited by none of them. So the gap this addresses is not a contested claim in need of correction; it is that a disease defined by a transcription-factor fusion has **no assembled account of what that fusion transcribes**, and the question is asked rarely enough that the distinction between the two above has had little occasion to be drawn. Query strings, counts and their limits are in the deposited probe artifact.

The first is a mechanism claim; the second is an association. A gene can satisfy the second for reasons that have nothing to do with the fusion — EMC's cell of origin, its myxoid and hypocellular architecture, the anatomical site it arises in, or the gene being a generic matrix or proliferation gene. This work separates the two by (a) cataloguing the mechanism claims with their evidence type recorded per gene, (b) reading them back in tumour tissue with an explicit calibration for what an arbitrary gene set does on the same platform, and (c) putting each gene through the specific confounds that could have manufactured it.

2 • Materials and Methods

Full method detail — probe mapping, scoring floors, the circularity grading procedure, the pre-registered decision rule and the motif-scan parameters — is in Supplementary Information §S1–S6.

2.1 • The evidence-typed target catalogue

Every claim in the primary literature that EWSR1::NR4A3, another NR4A3 fusion, or native NR4A3 transcriptionally activates a named gene was recorded with: the gene, the factor actually tested, the assays, the cell system, the species of those cells, the expected direction in EMC, and the verbatim sentence the classification rests on. Rows were read from retrieved full text, not from memory. Four evidence classes were assigned per row (**Table 1**; the complete 22-row catalogue with verbatim sentences is Supplementary Table S1).

Table 1. The evidence classes, and how many genes are in each.

Table 1		
class	definition	genes
A — fusion DNA-binding	a DNA-binding or promoter assay performed with an NR4A3 fusion . The strongest class.	3
B — native DNA-binding	the same assay class with native NR4A3 . Transfer to the fusion is an assumption.	16
C — fusion expression only	the gene moves when the fusion is expressed; no binding assay.	2
D — EMC tumour expression only	measured in EMC tissue; no mechanism.	1

2.2 • Datasets

Three independent EMC cohorts on three platform families were used. They are never pooled (§5).

Table 2. The three cohorts.

Table 2				
cohort	platform	EMC	comparators	value kind
GSE24369	GPL6244, Affymetrix Gene ST	6	29 — 17 FET-rearranged LGFMS + 6 myxofibrosarcoma + 6 desmoid fibromatosis	single-channel intensity
GSE4303	GPL3290, 42,000-spot two-colour cDNA	10	6 (3 DFSP + 3 GIST)	two-colour log-ratio vs a reference pool
GSE28866	3SEQ (GPL10999)	4	32 non-EMC sarcoma libraries; 27 normal-organ libraries	3'-end read density per peak

Every class label above is the GEO sample title, not an internal grouping name. That distinction decides §3.4: **23 of the 29 GPL6244 comparators are themselves myxoid tumours** — 17 low-grade *fibromyxoid* sarcomas and 6 *myxofibrosarcomas* — against 6 collagen-rich desmoid fibromatoses, whereas none of the 6 GPL3290 comparators is myxoid. The two array cohorts therefore differ in the one property the leading confound is built on, which makes the pair a natural control rather than only a limitation.

- **GSE24369** — its comparator arm is itself FET-rearranged (LGFMS is *FUS::CREB3L2*), so a difference here is not merely "has a FET fusion". The array carries 42 samples; 6 EMC + 29 comparators accounts for 35, and the remaining 7 — ****five solitary fibrous tumours and two pooled skeletal-muscle RNA samples**** — are excluded

from both arms rather than silently absorbed into either, so the arithmetic closes. The two muscle samples are named here because they are not merely surplus: they are the reference that makes the *ENO3* muscle-admixture objection measurable (§3.5). Linked series PubMed identifier 21536545.

- **GSE4303** — Subramanian *et al.*, *J Pathol* 2005;206:433–444 (PMID 15920699). See §3.8 on circularity. Two structural facts about this deposit bear on every number read from it. It is a **seven-platform series** — seven sibling print runs of one clone library — of which only GPL3290 carries a usable EMC-versus-comparator contrast, so the 10 versus 6 here is not the whole deposit. The **published** Subramanian cohort was 10 EMC against 26 other sarcomas, so a reader opening the accession will find comparators this analysis does not use. Separately, the verbatim sample annotations record the reference pool each two-colour hybridisation was run against: all 10 EMC and all 3 DFSP samples are on the CRH pool, while ****the 3 GIST samples are on Universal Human Reference — a different pool**** (§3.4).
- **GSE28866** — Brunner *et al.*, *Genome Biol* 2012;13(8):R75 (PMID 22929540). The EMC libraries are EMC_STT5525/5526/5527/5592; the normal arm is 27 libraries across six organs (bowel, breast, colon, kidney, lung, uterus). The 32 non-EMC sarcoma columns include two pairs of technical replicates of one specimen each (ESS_STT5520, LMS_STT516), so 32 libraries come from 30 specimens.

Each cohort's EMC arm size in Table 2 was read from its series matrix. All three were separately recovered from GEO **sample titles** — an independent path to the same numbers, 6, 10 and 4 — during the cohort search of §2.7, where they serve as its positive control: a search that cannot find the datasets that exist says nothing about the ones that do not.

△ One deposit is not a fourth cohort and is easy to mistake for one. **GSE170983** carries 99 samples, four of them EMC, under its own accession — and it is the same Brunner deposit as GSE28866, with the same four tumours (GSM715466/715467/715470/715472) and the same linked publication. Counting the two separately would raise the apparent EMC total to 24 without adding a patient.

2.3 · Scoring and the size-matched empirical null

Probes were mapped to symbols per platform (GPL6244: 20,230 of 28,459 probes → 18,694 distinct symbols; GPL3290: 27,203 of 43,008 probes → 14,932, through an EST-accession bridge). Each sample's values were z-scored against that array's own probe distribution; a gene or set score for a sample is the mean z over its readable members; the contrast is a Welch *t* on the EMC versus comparator per-sample scores. **A gene with no probe is treated as an unread gene, never as an absent one.** Floors were fixed at three samples per group for any contrast, and four genes / 0.4 coverage for any set score.

Two quantities a raw Welch contrast does not supply are computed per platform. **The exact global offset** is the per-sample mean z over every symbol the platform maps, contrasted EMC versus comparator — the amount by which an arbitrary gene set is expected to differ for no set-specific reason. **The size-matched empirical null** is 4,000 random gene sets of exactly the observed size, drawn from a seeded random pool of the platform's own mapped symbols (seed 20260807; pool 4,000; universe 18,694 for GPL6244, 14,932 for GPL3290), each scored exactly as the real set is. A random set carries the offset too, so the null absorbs it by construction. The empirical p is the **two-sided** fraction of draws at least as extreme, with +1/+1 smoothing. A set is reported as SET-SPECIFIC only if the observed delta falls outside the 95% band of that null; single genes are calibrated the same way at set size 1.

The empirical p has a floor, and it is quoted as one. With 4,000 draws and +1/+1 smoothing the smallest two-sided value the design can return is $2/4001 = 0.0005$, and every such value below is written **p_emp ≤ 0.0005** — the resolution limit of the null, not a measured value. The floor is not the same on both platforms. A draw is discarded when the gene it picks leaves either arm below the three-sample contrast floor, so every size-1 null on

GPL3290 retains 3,700 of its 4,000 draws and its floor there is $2/3701 = 0.00054$, the value Table 4 prints for *ENO3*. Three further limits of the pool are stated rather than left implied: it is a **seeded 4,000-symbol random subsample** of each platform's mapped symbols (21% of GPL6244's 18,694, 27% of GPL3290's 14,932), the same pool is reused at every set size, and the gene under test is not excluded from it.

The null's own limit is stated rather than assumed: it is a **competitive** null, controlling for the platform-wide offset and for set size, but not for gene–gene correlation inside a real pathway. It is therefore anti-conservative for coherent sets and is a screen, not a test — which is why §2.6 supplies a self-contained null alongside it.

2.4 • Instrument controls, graded before the biology

Four known answers were graded before any biological read. The first two are *ENO3* (UP on both platforms — the positive control) and *NR4A3* (UP — tumour identity). The other two are *PLAGL1* (DOWN, PMID 16112421 — the directional falsifier, the only prediction an arm-wide offset cannot manufacture) and *SGK1* (flat or down at transcript level despite 10/10 protein positivity, PMID 16756948 — the only row whose published transcript and protein directions oppose).

Grading is on where the delta sits relative to its size-1 null, never on the raw delta. **The rule has three outcomes, not two, and the third is the one that needs stating.** A reading whose delta falls *outside* its null band is graded, and agrees or disagrees with the published direction. A reading whose delta falls *inside* the band is **not a reading at this power** and is not graded either way — a randomly chosen gene on that platform would land there too. And a control with no computable contrast at all (for example, *NR4A3* on GPL3290, where four of six comparator spots for that probe are missing, leaving two values against a floor of three) is likewise not graded.

Two consequences follow, and both are reported rather than left for a reader to find. First, the denominator of any "n of n agree" statement is the number of **gradeable readings**, not the number of controls — §3.3 gives both. Second, a control can only fail by landing outside its band in the *wrong* direction, so a control with a wide band is weakly falsifiable; §3.3 therefore prints each control's band beside its verdict, because a prediction that could not have been refused is not evidence that it was tested. An absent reading is not a reading of absence, and the block whose job is to tell a working instrument from a broken one is the last place that distinction may collapse.

2.5 • The 3SEQ arm, and its calibration

3SEQ measures 3'-end read density per peak. A gene's value in a library is the median across that gene's peaks; an arm's value is the median across that arm's libraries. **No z-score, no test and no confidence interval are used, because n = 4.** 3SEQ read density is not array intensity, and nothing from this arm is pooled with the arrays.

A fold-change is not a reading until an arbitrary gene's fold-change is known, so the same two ratios were computed for **every gene in the deposit** (14,120 genes; 13,708 with a computable EMC/normal ratio, 13,247 with an EMC/sarcoma ratio) and each target gene is reported as a **percentile of that distribution**. A gene whose comparator median is zero has no ratio and is excluded from the distribution rather than ranked at the top. A percentile is a rank, not a p-value, and none is implied.

2.6 • Confound audit and sensitivity analyses

Four further readings were computed offline from the same cached inputs, all using the same scoring primitives so the quantity tested is identical to the quantity reported. Every delta re-derived here was asserted equal to the primary artifact before anything was written.

1. **Exact sample-label permutation** (a self-contained null). The EMC/comparator label is permuted over samples and the *real* gene set rescored, so correlation structure is carried through untouched. Arm sizes give 1,623,160

and 8,008 distinct assignments — few enough to **enumerate completely**, so every reported p is exact rather than sampled. Two-sided throughout.

2. **Every comparator stratum, separately.** The contrast is recomputed with the comparator arm restricted to one class at a time, to the myxoid classes only, to the non-myxoid classes only, and — on GPL3290 — to the **reference-pool-matched** comparators only. Nothing about the EMC arm or the reduction changes, so any movement is attributable to who is in the comparator arm. Each stratified contrast carries its own exact permutation p ($C(29,6) = 475,020$ down to $C(13,10) = 286$, all enumerated completely).
3. **Covariate-adjusted sensitivity.** Each gene's per-sample z is regressed on a matrix-content proxy and the contrast recomputed on the residuals. **The proxy is selected by provenance:** every candidate is checked against every gene this manuscript scores anywhere, and any gene drawn from an EMC-derived list is refused, because adjusting on a gene selected *because* it is high in EMC removes EMC signal by construction. Two panels survive that filter: 11 structural genes (*BGN*, *COL5A1*, *COL5A2*, *DCN*, *FN1*, *LUM*, *MMP2*, *POSTN*, *SPP1*, *TNC*, *VIM*) and 3 vascular genes (**ENG*, *KDR*, *TEK**). The structural panel is the one §3.6 reports first; the vascular panel's reading is reported beside it, because the two do not agree on GPL3290. This is a sensitivity analysis, not a correction: a proxy that is itself downstream of the fusion would over-adjust, and that possibility is not excluded.
4. **The skeletal-muscle admixture control** (§3.5), plus a **leave-one-out jackknife** over the EMC arm, a **rank-based re-read** on within-array percentile, and **Benjamini–Hochberg** q-values across the per-gene permutation p-values within each platform.
5. **NR4A occupancy** (§3.11). The scan intersected 110 published NR4A ChIP-seq peak sets — ChIP-Atlas, ReMap2022 and the Haller *et al.* acinic cell carcinoma deposit — with the class-A genes' regulatory windows, the same window as the motif scan, so the sequence and occupancy axes ask about one region. Every count was placed against a background panel of 198 genes assembled for an unrelated question. Four rules govern the reading. A **raw count is never reported as a finding**, because the deepest catalogue puts a peak in 82.8% of the panel. A peak set that recovers (almost) no panel gene is marked **uninformative**: it cannot fail to recover these three, so its silence is an absent reading and is never counted as evidence of non-occupancy. **Only NR4A antigens are scored** — the Haller deposit also carries CTCF, H3K27ac, H3K27me3, H3K4me3 and super-enhancer calls, and a histone peak at a promoter reports that the promoter is active, not that an NR4A protein is there. And the nominal-hit count is judged by a **binomial tail** against the number of tests rather than by comparing an integer to a fractional expectation. Multiplicity is over distinct **experiments**, not genome builds, since the same experiment appears once per build.

The Haller peak files carry no genome build, and a BED intersected on an assumed build does not fail — on chr10 it silently reports another locus. The build was therefore **measured**: H3K4me3 marks active promoters, so on the correct build it must recover most of the same background panel and on the wrong one it must not. All four samples independently gave **90.6–93.9% panel recovery on hg19 against 32.2–33.6% on hg38**, and the deposit is read as hg19. Both an absolute floor (0.80) and a ratio (2.0) are required, because the two builds agree over much of the genome, so ~33% is the expected wrong-build floor rather than noise.

2.7 • The cohort search

Six deliberately overlapping queries were put to GEO — the disease name in full and abbreviated, the fusion rather than the disease, the 3' partner plus lineage, and two over-broad terms chosen to return pan-sarcoma and general chondrosarcoma deposits in which EMC samples might sit under a generic title. Every query is recorded with its own stated purpose, including any returning nothing, because a query that returns nothing is indistinguishable from a dataset that does not exist unless the query itself is on the record.

Three properties of the procedure matter more than the queries. **Every returned series is read at sample level**, not screened on its title: GEO titles are depositors' claims, and GSE24369 — titled after low-grade fibromyxoid sarcoma while carrying six EMC tumours — is the standing demonstration that a title-level screen would discard the very cohorts this analysis rests on. **Every candidate is deduplicated at three levels** — accession, linked publication, and sample identity against all 157 GSM records the three cohorts read — because a re-deposit under a new accession is otherwise indistinguishable from a new cohort. And **the search is graded against a positive control**: the three cohorts already in use must be recovered by the same queries, with their EMC arm sizes, or the result is withheld as uninterpretable rather than reported as a negative.

2.8 · Reproduction

Every value in §3 derives from a committed artifact and is not re-typed from prose. All analyses are CPU-only, use open-source tooling, and reproduce offline from cached inputs with no network access; producers refuse to write if any row disagrees with the artifact that owns it. See Data and code availability.

3 · Results

3.1 · Three genes are the whole of class A

Table 3. The complete class-A catalogue — every gene for which an NR4A3 chimera has been shown to bind DNA.

gene	chimera assayed	assays	cells	citation
SEMA3C	EWSR1::NR4A3 (and TAF15::NR4A3, and native)	predicted NBRE-like site (GRCh38 chr7) + ChAP-qPCR, Strep-tagged	tBJ/ER transformed human fibroblasts	Brenca <i>et al.</i> , <i>J Pathol</i> 2019;249(1):90–101 (PMID 31020999)
PPARG	EWSR1::NR4A3 (and native, and NR4A3ΔC)	predicted perfect NBRE at –675 bp, band-shift, 2.8 kb human <i>PPARG</i> promoter luciferase, single-nucleotide NBRE mutant	CFK2 fetal rat chondrogenic cells; human promoter construct	Filion <i>et al.</i> , <i>J Pathol</i> 2009;217(1):83–93 (PMID 18855877)
ENO3 (β-enolase)	TFG::NR4A3 — <i>not</i> EWSR1	EMSA + ChIP + luciferase, two NBRE motifs upstream of the TSS, plus ChIP for H3 acetylation at the endogenous promoter	cultured lines over-expressing TFG-TEC	Kim <i>et al.</i> , <i>Mol Carcinog</i> 2016 (PMID 26310886)

Three genes are the whole of class A, and this is the most consequential number in the report (**Figure 2**). Across the retrieved corpus, the number of genes anyone has shown an NR4A3 chimera physically binding and driving is three — and only one of the three (*SEMA3C*) combines the EWSR1 chimera, human cells and a chromatin assay. Three is the count in 2,276 retrieved full-text documents across five corpora (§3.11), not a claim about all of the literature.

Class B — the same assay class with native NR4A3 — holds sixteen genes: *CCND1*, *SKP2*, *VTN*, *SMPX*, *CDKN2AIP*, *GLS2*, *SDHA*, *COX5A*, *PDP1*, *VCAM1*, *ICAM1*, *BIRC3*, *NOX1*, *TH*, *LOXL2*, *MYH7*. The strongest are *SMPX* (promoter deletion + site-directed mutagenesis + EMSA + ChIP, human cells, PMID 27181368), *SKP2* (EMSA + ChIP) and *CDKN2AIP* (ChIP + mutation-reversed reporter, human cells, PMID 39664575). **Class C** holds *SGK1* and *PLAGL1*; **class D** holds *NDRG2*. A published negative control is carried alongside them: *CALD1*, whose promoter was searched for NOR-1 response elements in the same experiment that found the *SMPX* site, and none were found (PMID 27181368). It controls the inference "this gene moved, therefore NR4A3 bound it", not EMC biology.

3.2 • The native → fusion transfer assumption is measured to fail in both directions

Class B is only usable if a native-NR4A3 target is a fusion target. Two published measurements say the transfer can fail, in opposite directions:

1. **A native target the fusion does not share.** Filion *et al.* put native NR4A3 and NR4A3ΔC on the same *PPARG* reporter the fusion activates: "the results show that both the native and truncated receptors do not activate *PPARG* transcription under the same conditions in which it is readily activated by the fusion protein."
2. **A fusion target the other fusion does not share.** Brenca *et al.*: "the ability of NR4A3 to recognize the SEMA3C target region was retained by the EWSR1-NR4A3 chimera but was impaired by TAF15-NR4A3."

So "NR4A3 binds X" does not license "EWSR1::NR4A3 drives X in EMC", and a native-NR4A3 cistrome is not a fusion cistrome. Both halves of that are demonstrated in the primary literature, not argued here.

3.3 • The instrument controls

Table 4. The four instrument controls, graded before any biological read.

Table 4		
control	GPL6244	GPL3290
<i>ENO3</i> (positive)	AGREES — d +0.8075, outside null, p_emp 0.0195	AGREES — d +3.8113, outside null, p_emp 0.00054
<i>NR4A3</i> (tumour identity)	AGREES — d +0.7415, outside null, p_emp 0.0240	not measurable — 2 comparator values against a floor of 3
<i>PLAGL1</i> (directional falsifier)	inside null, not a reading at this power — d -0.4235, band [-0.606, +0.529]	AGREES — d -2.134, outside null, p_emp 0.013
<i>SGK1</i> (transcript/protein discordance)	AGREES (flat) — d -0.1807, band [-0.606, +0.529]	AGREES (flat) — d +0.6156, band [-1.314, +1.410]

Seven of the eight control × platform cells carried a computable contrast, six of those are gradeable, and all six agree with the published direction; none disagrees. The eighth (*NR4A3* on GPL3290) is not measurable. Stated at the weight it deserves: *PLAGL1* on GPL6244 is **inside its null band** and is therefore sign-concordant but not a reading at this power. The two *SGK1* cells agree by way of a prediction ("flat or down") that an inside-the-band reading satisfies, so those cells could not have refused the prediction downward, and their bands are printed above for that reason.

The positive control is independently reproduced: *ENO3* matches a separately written module's committed value to four decimal places on both platforms. The directional falsifier fires in the right direction where it is gradeable: *PLAGL1*, the one gene in the catalogue with a published *down* prediction, is -2.13 SD in EMC on GPL3290 and outside its null band, while every other class-A row points up; no arm-wide artefact produces that pattern. Its published EMC reading is n = 6 by RT-PCR against chondrocyte controls rather than sarcomas, and its fusion-expression evidence is differential display in rat cells, so it is a strong argument against the offset explanation rather than a fully independent falsifier.

3.4 • The global offset is not the problem — null-band width is

The measured global offset is tiny: -0.0084 SD on GPL6244 (t -1.592, over 18,694 mapped symbols) and +0.0258 SD on GPL3290 (t +1.646, over 14,932). So the "most sets come back higher in EMC" pattern is not an arm-wide shift. What it is instead: at n = 6 versus 29 and n = 10 versus 6, the sampling variance of a set score is far larger than a Welch *t* on the sample means implies. On GPL3290 the 95% null band for the 17 readable members of that set is [-0.297, +0.376], so a raw delta of +0.330 with t = 3.16 sits inside it (p_emp 0.083). **Figure 1** shows this directly.

Two structural properties of the comparator arms qualify every contrast below, and both are exploited rather than merely conceded. The GPL6244 comparator arm is **23/29 myxoid**, so it largely matches EMC on the matrix property that confound (b) of §4.1 is built on; the GPL3290 comparator arm is **0/6 myxoid**, so it does not. And the GPL3290 comparator arm is split across two reference pools — 3 DFSP on CRH with all 10 EMC samples, 3 GIST on Universal Human Reference — which is a per-gene offset that within-sample standardisation cannot remove, because standardisation removes a sample's mean and SD, not a per-gene shift. §3.6 recomputes every class-A contrast against the pool-matched comparators alone for that reason.

3.5 · Per gene, and what survives

Figure 3 shows every tumour. Figure 4 summarises which instrument supports which gene.

Table 5. The three class-A genes on both array platforms, under an exact label-permutation test.

Table 5 · numbered notes are under the caption, on this table's first page			
gene	class	GPL6244 Δ mean z (exact p, BH q)	GPL3290 Δ mean z (exact p, BH q)
ENO3	A · fusion	+0.8075 (7.3 × 10 ⁻⁵ , q 0.00044)	+3.8113 (1.3 × 10 ⁻⁴ , q 0.00063)
PPARG	A · fusion	+0.3071 (0.049, q 0.097)	+2.4809 (3.3 × 10 ⁻⁴ , q 0.00083) — circular, §3.8
SEMA3C	A · fusion	+0.7298 (0.194, q 0.233)	+0.6228 (0.165, q 0.165)

All three genes are positive-signed on both platforms — six of six readings, no reversal **against the pooled comparator arm**, and each clears its size-matched single-gene null on at least one. That qualifier is load-bearing: §3.6 shows *SEMA3C* reversing sign once the comparator arm is taken apart, so "no reversal" is a property of one particular way of pooling the comparators and not of the gene. Sign concordance across three genes is in any case what a coordinated programme predicts *and* what three individually EMC-associated genes predict, and the three are not equally supported once the self-contained null is applied. Under exact sample-label permutation, **ENO3 is significant on both platforms after multiple-testing correction**, *PPARG* on GPL3290 only — which §3.8 shows is the circular platform — and **SEMA3C does not reach significance on either**. Clearing the size-matched null says a gene's delta is extreme relative to *other genes on the platform*; it is not the same statement as the two arms differing for that gene. No row in the panel changed sign in any leave-one-out fit, and none changed sign on the rank re-read, so nothing here rests on one tumour or on the z-scoring convention.

ENO3 is also this study's positive control, and that has to be said plainly. §2.4 designates it the control whose failure would mean "report the instrument, not the biology" — so its elevation in EMC is not an independent finding of this work. It was chosen as the control *because* Kim *et al.* published it as fusion-driven and a separately written module had already committed its value on both platforms. Two things keep the rest of the reading from being circular, and a reader should weigh them rather than take the word "survives" at face value. First, the control role tested one proposition only — is it up on both platforms — and **everything that separates ENO3 from PPARG and SEMA3C here was not part of it**. The exact permutation p, invariance across five comparator strata, the reference-pool-matched contrast, the matrix adjustment, the 3SEQ percentile, the muscle control and the NBRE enrichment could each have failed and did not. Second, the finding this paper reports is the **ordering** of the three genes, not *ENO3's* elevation, and an ordering cannot be manufactured by having selected one member in advance. What remains true regardless is Limitation 17: a design in which the positive control and the surviving result are the same gene is structurally weaker than one in which they are not, and only an independent gene reaching the same bar would remove that.

The muscle-admixture objection, and its answer (Figure 5). *ENO3* is muscle-specific β-enolase and EMC arises in deep soft tissue of the limb, so admixed skeletal muscle is the first alternative explanation a reader should reach for. GSE24369 contains two pooled skeletal-muscle RNA samples, in neither arm and used by no contrast, which fix the scale of what muscle looks like on this platform. *ENO3* does sit near the top of the muscle array (percentile

0.996). **So do four markers that are more muscle-restricted than it is — *ACTA1* 1.000, *MYH7* 1.000, *PYGM* 0.999, *MYL1* 0.998 — and not one of them separates the tumour arms as *ENO3* does** (EMC – comparator –0.057, –0.043, +0.142, –0.150 percentile points, against *ENO3*'s +0.315). *PYGM*'s +0.142 is the largest of the four and is under half *ENO3*'s. If the EMC arm carried skeletal muscle, the more muscle-restricted markers would carry it too. This bounds admixture of *differentiated* skeletal muscle; it does not exclude a myogenic differentiation programme within the tumour, which would move a marker with no contaminating tissue present.

3.6 · The contrast against every comparator stratum separately

The comparator arm is not one thing, and the class-A genes behave very differently when it is taken apart. Each cell is the same contrast with the same EMC arm, its own exact permutation p in brackets.

Table 6. The same contrast against each comparator sub-arm separately.

Table 6 · numbered notes are under the caption, on this table's first page				
gene	vs LGFMS only (17)	vs myxoid only (23)	vs non-myxoid only (6)	GPL3290, pool-matched only (3)
ENO3	+0.805 (1.7×10^{-4})	+0.808 (8×10^{-5})	+0.807 (0.022)	+3.515 (0.0035, the design floor)
PPARG	+0.197 (0.220)	+0.264 (0.100)	+0.473 (0.043)	+2.679 (0.0035) — circular
SEMA3C	+1.657 (1.2×10^{-4})	+1.089 (0.046)	–0.645 (0.015)	+0.113 (0.843)
<i>PLAGL1</i>	–0.169 (0.431)	–0.343 (0.183)	–0.733 (0.043)	–1.659 (0.042)

***ENO3* is invariant** — +0.805, +0.808, +0.807 across strata that share almost nothing, and significant against every one of them, including the myxoid-matched arm that controls confound (b) by design. ***SEMA3C* reverses sign:** it is strongly up against LGFMS and significantly *down* against desmoid fibromatosis, and on GPL3290 it is +0.113 (p = 0.84) against the pool-matched comparators. Its apparent elevation is a property of which sarcomas happen to be in the comparator arm, not of EMC. *PPARG* is intermediate and comparator-dependent.

Two readings of this table are worth stating explicitly, because both are easy to miss. First, *SEMA3C* is *not* significant against the pooled comparator arm (p = 0.194, Table 5) yet is significant against two of its strata in **opposite directions**. That is not a contradiction: the pooled arm averages a stratum where *SEMA3C* is low (LGFMS) with strata where it is high, and EMC lands between them, so two strong opposite effects cancel into a null. A pooled contrast can therefore conceal large stratum effects, and a gene that looks flat against a heterogeneous comparator arm has not been shown to be flat. Second, and conversely, the stratified panel is where a gene can most easily be flattered: with five contrasts per gene on GPL6244 and no correction across them (Limitation 8), the least favourable stratum is the honest summary, which is what **Figure 4** reports. On that measure *ENO3* is significant at its worst stratum (p = 0.022) and *SEMA3C* is not (p = 0.136 at its worst).

The reference-pool correction matters and does not overturn *ENO3*: restricting GPL3290 to the three pool-matched DFSP comparators moves it from +3.811 to +3.515, still the smallest p that a three-comparator design can report (1/286 = 0.0035, quoted as the floor it is).

Covariate-adjusted sensitivity. The 11-gene matrix panel separates the arms on GPL6244 (Δ –0.518; EMC scores *lower*) and essentially not at all on GPL3290 (Δ +0.006). That asymmetry is the method's own control: a covariate that does not differ between the arms cannot move a contrast, and on GPL3290 nothing moves (*ENO3* retains 100%, *PPARG* 97%, *SEMA3C* 99%). On GPL6244, where the covariate does differ, *ENO3* retains 75% of its delta (+0.807 → +0.608), *PPARG* 32% (+0.307 → +0.099), and *SEMA3C* rises to 171% (+0.730 → +1.246) — it is high in EMC *despite* EMC's lower matrix score, which is consistent with its elevation being comparator-driven rather than matrix-driven.

The second surviving panel does not agree, and it is reported rather than left out. The 3-gene vascular panel (*ENG*, *KDR*, *TEK*) passed the same provenance filter, and unlike the matrix panel it separates the arms on **both**

platforms ($\Delta -0.406$ on GPL6244, $\Delta -0.796$ on GPL3290), so on GPL3290 it can move a contrast where the matrix panel cannot. Adjusted on it, *ENO3* retains **58%** of its GPL3290 delta ($+3.811 \rightarrow +2.219$) and 78% of its GPL6244 delta; *PPARG* retains 42% and 30%; *SEMA3C* 85% and 189%. So *ENO3*'s GPL3290 reading is not invariant to which qualified covariate is used, and the 100% above is a property of the matrix panel rather than of the gene. The ordering of the three is unchanged on either panel.

3.7 · A third cohort, calibrated against its own deposit

The 3SEQ arm carries both contrasts in one experiment: 32 non-EMC sarcoma libraries on an unrelated technology, and 27 normal-organ libraries. The median gene in this deposit has an EMC/normal ratio of 1.05 and an EMC/sarcoma ratio of 1.05; the 95th percentiles are 1.89 and 1.89.

Table 7. The 3SEQ cohort, calibrated against all 14,120 genes in the same deposit.

Table 7					
gene	peaks	EMC/normal	percentile	EMC/sarcoma	percentile
<i>ENO3</i>	2	2.53×	98.0th	2.02×	95.9th
<i>SEMA3C</i>	3	1.82×	94.2nd	1.66×	92.6th
<i>PPARG</i>	5	1.42×	84.0th	2.12×	96.4th
<i>NR4A3 (control)</i>	3	1.96×	95.6th	— (sarcoma median 0.000)	—

All three class-A genes are higher in EMC than in both comparator arms of a cohort and a technology that share no probe design with either array, and *ENO3* is in the top 2% of 14,120 genes on the normal axis. **The ceiling is that "top 2%" is not "highest":** *RET* (3.51×, 99.1st), *VCAN* (3.33×, 99.0th) and *CSPG4* (3.31×, 99.0th) all rank above *ENO3* in this deposit, so a high percentile here places a gene in the upper tail of an EMC-versus-comparator distribution and does nothing more. *PPARG* at the 84th percentile against normals is the weakest cell in the table and is the honest reading of its 1.42×

NR4A3 is the internal control here and is not a result: its median across the 32 non-EMC sarcoma libraries is 0.000 and it is detected in the EMC arm — the fusion's own 3' partner behaving as the disease definition requires, in a cohort this work did not choose and on an assay it did not design. It licenses reading the other rows; read as a finding it would be circular, because EMC is *defined* by an *NR4A3* rearrangement. The normal arm is a tissue panel, not matched adjacent tissue: the 27 normals are visceral organs with almost no soft tissue, so a gene high in EMC against that panel is not thereby EMC-specific rather than mesenchymal-lineage-specific.

3.8 · The circularity flag fired twice

The fetched GEO record for GSE4303 reads "Gene expression profile of extraskeletal myxoid chondrosarcoma", with linked PubMed identifier 15920699 and contributor "Matt van de Rijn". **GSE4303 is the Subramanian *et al.* (2005) cohort.** Two consequences, and the second was not drawn in the first version of this analysis:

1. The Filion Table 2 gene set (set E, Supplementary Table S2) is a gene list scored on the data it was derived from, and is reported for completeness only.
2. **The *PPARG* gene row on GPL3290 is circular in the same sense.** Subramanian *et al.* reported, from these arrays: *"High levels of expression of PPARG and the gene encoding its interacting protein, PPARGC1A, in most EMCs."* Measuring *PPARG* high in GSE4303 therefore re-derives a published finding from the data it was published from. It is not independent evidence, and a circularity grade applied to a gene set but not to a gene is not a grade. With that cell set aside, *PPARG*'s remaining evidence is GPL6244 ($q = 0.097$, which does not survive correction) and the 3SEQ cohort.

Set D and the whole of GPL6244 are unaffected, which is why the replication in §3.9 stands.

3.9 • The aggregate does not clear its null; the published EMC phenotype clears it 12-fold

Table 8. Gene-set scores against their own size-matched nulls, with the threshold each had to clear.

Table 8		
set	GPL6244	GPL3290
A • fusion DNA-binding targets (3)	no score — 3 genes, floor is 4	no score — 3 genes
B • native NR4A3 DNA-binding targets (16)	d -0.0675, reached 43% of threshold → not distinguishable	d -0.1453, reached 43% → not distinguishable
A+B pooled (19)	d +0.0403, reached 39% → not distinguishable	d +0.3301, reached 88% → not distinguishable
D • published EMC phenotype — EMC vs 137 other translocation sarcomas, independent platform and cohort (21)	d +1.1311, $p_{\text{emp}} \leq 0.0005$, 11.9× threshold → SET-SPECIFIC UP	d +1.4783, $p_{\text{emp}} \leq 0.0005$, 4.2× threshold → SET-SPECIFIC UP

This is the informative shape, and the "reached X%" column is what makes the negative interpretable rather than a shrug. The published EMC transcriptional phenotype — a gene list derived on a platform used nowhere in this work (Affymetrix U133A) from a cohort used nowhere in this work (MSKCC) — replicates cross-platform and cross-cohort at the null's resolution floor on both readable series, overshooting its threshold **11.9-fold and 4.2-fold**. On the same instrument, in the same run, the aggregate direct-target set reaches 39% and 88% of the threshold it would have had to clear. So the contrast demonstrably *can* see EMC at very large margin and does not see the aggregate target set — which is a bounded negative, not an underpowered one.

The native-NR4A3 set behaves as Filion's own measurement predicts: class B is flat-to-negative on both platforms, with *VCAM1* significantly down on both. The vascular/inflammatory native-NOR-1 programme does not transfer to EMC tissue — concordant with the same paper's finding that native NR4A3 does not activate the promoter the fusion does (§3.2). Set D is a test of the EMC phenotype, not of the fusion: a gene can be in it because of EMC's cell of origin, so its replication says the instrument reads EMC, not that the fusion drives those genes.

One result becomes more interesting rather than weaker. The A+B aggregate does *not* beat an arbitrary set of the same size on either platform — yet on GPL3290 it *does* differ between the arms more than chance relabelling would give (exact $p = 0.011$). That is not a contradiction; it is the cleanest available demonstration of this paper's own thesis. The aggregate target set really is higher in EMC, **and so is almost any set of that size on that platform**, which is precisely why a competitive null is the one that decides whether a gene set has told you anything.

3.10 • A sequence axis, and its limits

NR4A3's monomer site is the NBRE, 5'-AAAGGTCA-3' (PMID 1902986); the dimer site is the NurRE (PMID 9315667). A TSS-centred window of -10 kb/+15 kb — fixed in advance on published regulatory architecture, before any sequence was read — was scanned on both strands with positional de-duplication, and calibrated against a dinucleotide-preserving shuffle of the same window (2,000 shuffles, holding GC and CpG content exactly) and a 198-window background panel assembled for an unrelated question. Full parameters and the one-mismatch analysis are in Supplementary §S6.

ENO3 carries 4 exact NBREs, more than its own composition predicts (shuffle-null $p = 0.034$; GC-matched background $p = 0.018$). **PPARG carries 3**, which is what composition predicts ($p = 0.227$). **SEMA3C carries none**. The background panel averages 1.15 exact NBREs per 25 kb window, so a single hit is what an arbitrary window contains anyway.

Four things this does not establish. **A motif is not occupancy**, and the chromatin experiment that would test it now exists, was run, and does not support these sites. Frenkel *et al.* expressed EWSR1-NR4A3, TAF15-NR4A3, TCF12-NR4A3 and TFG-NR4A3 in HEK293T inside a pooled variant library and resolved each one's effect on chromatin accessibility by single-cell ATAC (GSE243553; PMID 39048711). Their per-fusion accessibility calls were

intersected with the four exact NBRE coordinates above on the matching genome build — the deposit declares hg38 and these coordinates are GRCh38, so no lift-over was required. **Three of the four sites fall inside TAF15-NR4A3's called intervals and none falls inside EWSR1-NR4A3's.** Neither observation supports the motif argument. The EWSR1-NR4A3 set is too sparse for its zero to be a reading: it recovers 2 of 203 promoters in a background gene panel assembled for an unrelated question, so a chosen gene could not have been recovered either. The TAF15-NR4A3 co-location does not clear a null that slides the same four-site configuration, at its true spacing, to a random offset within the same window ($p = 0.08$, 20,000 seeded draws). That null is the one that matters here because two of the four sites are 153 bp apart and a single 500 bp interval covers both; nulls treating them as independent return $p \leq 0.002$ and overstate the result by a factor of about forty. The TFG-NR4A3 arm — the fusion Kim *et al.* used for the published *ENO3* result — is likewise too sparse to grade. 🚫 Three conditions bound all of this: the cells are HEK293T and not EMC chromatin, accessibility is not binding, and *ENO3* is this study's own pre-designated positive control (§2.4), so a hit there would have been a check on the instrument rather than a discovery. **The sequence axis and the chromatin axis have now been made to meet at these coordinates, and they do not corroborate each other.** Method, seeds and the four nulls: [gse243553-eno3-overlap-2026-08-08.md](#). **The SEMA3C zero does not contradict Brenca *et al.***, who report a predicted NBRE-like site assayed by ChAP-qPCR; an NBRE-like site is by construction not an exact NBRE. That class was therefore scanned too, and *SEMA3C*'s 39 one-mismatch sites — the most of any gene scanned — are **exactly what its own composition predicts** (null mean 33.7, $p = 0.203$; GC-matched $p = 0.118$), with only the composition-naïve raw rank suggesting enrichment ($p = 0.040$) in the most AT-rich window of the set. **The hit positions do not reproduce the published coordinates** for either *ENO3* or *PPARG*, both of which numbered from their own promoter constructs. **A distal element outside the window is untested by construction** — while the elements *inside* the window have now been tested against an orthogonal instrument, and were not corroborated by it.

3.11 · Nothing has been measured in EMC chromatin — a bounded negative about a search

The obvious discriminator between *driving* and *correlation* is a cistrome, so five corpora totalling **2,276 full-text documents** (3,669 catalogued Europe PMC records) were searched. 153 of those documents name both a genome-wide chromatin method (ChIP-seq, CUT&RUN, CUT&Tag, ChIP-exo, ChIP-PET, ATAC-seq, ChAP) and NR4A3/NOR-1/TEC. **None of the 153 applies one to an NR4A3 chimera**, and the only chromatin experiment performed with a fusion anywhere in that corpus is Brenca *et al.*'s ChAP-qPCR — target-specific amplification at one locus, not a genome-wide map.

⚠️ **That count is a fact about a literature screen, and the absence it was read as does not follow from it.** A wider search on 2026-08-08 — the primary sequence archives rather than the literature alone, 179 API endpoints across six rounds with every query string committed — retrieved a genome-wide chromatin experiment performed with NR4A3 fusions. **GEO GSE243553** (Frenkel *et al.*, PMID 39048711; public 2024-07-24) is a pooled single-cell ATAC screen of more than 100 oncofusions expressed in HEK293T, and its 116-member variant library carries **EWSR1-NR4A3, TAF15-NR4A3, TCF12-NR4A3 and TFG-NR4A3** together with two controls this paper's argument needs: **full-length wild-type NR4A3**, and the reciprocal **NR4A3-EWSR1**. Quoted from that paper and not re-derived here: TAF15-NR4A3 increased accessibility at $\approx 8,600$ peaks, within which the NR4A-family motif was enriched; EWSR1-NR4A3 gave 1,235 differentially accessible peaks across 112 nuclei; full-length wild-type NR4A3 changed 0 peaks; and the reciprocal NR4A3-EWSR1 gave 0 peaks over 503 nuclei. **Why a screen of retrieved full text could not reach it is worth recording, because the shape recurs.** In a pooled screen the perturbation identity is data rather than metadata, so NR4A3 appears zero times in that paper's abstract and zero times across all 24 of the series' GEO sample records. This project's prior chromatin census was also antigen-centric with a ChIP-seq-only method vocabulary, which no ATAC deposit can satisfy. Search record, per-query counts and the retrieval-failure ledger: [nr4a3-cistrome-search-2026-08-08.md](#).

— **It is accessibility and not occupancy, HEK293T and not EMC, ectopic and not endogenous**, so it is not a fusion cistrome and must never be cited as one. What is missing is therefore narrower than a blanket absence, and can be stated exactly: **no experiment has measured where an NR4A3 fusion binds, or what chromatin does, in EMC material**. Across GEO, SRA, BioProject, BioSample, ArrayExpress/BioStudies, ENA and ChIP-Atlas, searched on 2026-08-08, an EMC disease term returns zero deposits carrying any chromatin library strategy. The 46 SRA runs an EMC term does return are every one RNA-Seq, WXS, WGS, Targeted-Capture or CAGE. ChIP-Atlas's complete antigen index carries NR4A3 in one cell type only (CD1c⁺ dendritic cells) and EWSR1 in seven, none of them EMC. **The negative is sharpest stated comparatively, because the field runs this experiment routinely for the sibling fusions and has never run it here** — ChIP-seq for EWSR1::WT1 and for EWSR1::ATF1, ATAC-seq for EWSR1::FLI1 and FUS::DDIT3 (GSE235218), and ChIP-seq twice for HEY1::NCOA2 mesenchymal chondrosarcoma (GSE163585, GSE196000). Δ It remains a statement about what has been deposited under a label an archive indexes, not about what exists: GSE243553 was itself invisible to every gene-keyed query in that sweep and was reached only through a paper's full text. Within that bound, a fusion cistrome in EMC chromatin is an open, unclaimed experiment rather than a dataset someone forgot to fetch.

The available surrogates were then measured rather than dismissed, because "no fusion cistrome" invites the reasonable objection that *some* NR4A chromatin data exists and might substitute for one. All 110 NR4A peak sets — from ChIP-Atlas, ReMap2022, and the Haller *et al.* acinic cell carcinoma deposit described below — were intersected with the class-A genes' regulatory windows, the same -10 kb/ $+15$ kb window as the motif scan, so the two axes ask about one region. Each count was placed against a background panel of 198 genes assembled for an unrelated question (**Table 9**).

Deep NR4A3 occupancy in human tissue does exist, in another disease. Acinic cell carcinoma of the salivary gland activates *native* NR4A3 by enhancer hijacking, and Haller *et al.* mapped it: NR4A3 ChIP-seq in three carcinomas and one normal parotid gland at 8,501–18,666 peaks each (Zenodo 10.5281/zenodo.1483691). That is 55–121 $\times$ the deepest NR4A3 peak set otherwise available, and it is the only NR4A3 cistrome in human tissue this analysis could reach. Δ **It is not a fusion:** these tumours carry over-expressed wild-type NR4A3, not EWSR1::NR4A3, and §3.2 records native NR4A3 failing to activate the *PPARG* promoter the fusion activates. It answers where the NR4A3 DNA-binding domain goes in a human tumour — a fourth axis of evidence — and it is never a substitute for the missing experiment.

Table 9. NR4A occupancy at the class-A genes, calibrated against a 198-gene background panel. Peak counts are promoter-window peaks; *p* is empirical against the panel.

Table 9						
experiment	antigen	peaks	panel genes with a peak	<i>ENO3</i>	<i>PPARG</i>	<i>SEMA3C</i>
ReMap2022 (merged)	NR4A1	83,773	82.8%	6, <i>p</i> 0.14	1, <i>p</i> 0.83	1, <i>p</i> 0.83
SRX1653204	NR4A1	26,660	45.5%	2, <i>p</i> 0.12	0, <i>p</i> 1.00	1, <i>p</i> 0.46
SRX1653203	NR4A1	22,717	31.3%	2, <i>p</i> 0.050	0, <i>p</i> 1.00	1, <i>p</i> 0.32
AciCC-1 (Haller)	NR4A3	18,666	67.5%	4, <i>p</i> 0.070	0, <i>p</i> 1.00	1, <i>p</i> 0.68
AciCC-2 (Haller)	NR4A3	9,810	56.0%	3, <i>p</i> 0.094	0, <i>p</i> 1.00	0, <i>p</i> 1.00
AciCC-3 (Haller)	NR4A3	9,263	49.0%	2, <i>p</i> 0.16	0, <i>p</i> 1.00	0, <i>p</i> 1.00
Normal parotid gland (Haller)	NR4A3	8,501	50.5%	4, <i>p</i> 0.035	0, <i>p</i> 1.00	0, <i>p</i> 1.00
5 further NR4A1 experiments	NR4A1	305–16,023	2.5–27.3%	0–1, <i>p</i> $\geq$ 0.26	0, <i>p</i> 1.00	0, <i>p</i> 1.00
12 ChIP-Atlas NR4A3 peak sets	NR4A3	53–154	0.0%	uninformative	uninformative	uninformative

The first number to read is the panel column, not the gene columns. In the deepest catalogue **82.8% of arbitrary genes carry a promoter-window peak**, so "has an NR4A1 peak" is what almost every gene does, and a raw count would be the same uncalibrated reading §1.1 exists to refuse. Across the 12 informative experiments, **2 of**

36 gene-by-experiment tests reach $p < 0.05$ against 1.8 expected by chance — a binomial p of 0.54 for that many or more, which is what chance routinely gives. No class-A gene carries unusual NR4A occupancy.

Three readings, in descending order of what they support.

- **PPARG carries zero promoter-window peaks in all four deep NR4A3 experiments** ($p = 1.00$ each). This is a **negative, not an absent reading**: those experiments recover 49–68% of the background panel, so they can find an arbitrary gene and did not find this one. It stands against Filion *et al.*'s perfect NBRE at –675 bp, band shift, and NBRE-mutant luciferase (Table 3). The two are reconcilable — a promoter can be bound by an over-expressed factor in a reporter assay and unbound in a different lineage's chromatin — but the tension is real and is reported rather than resolved.
- **SEMA3C carries at most one peak in one experiment**, consistent with every other axis on which it fails.
- *****ENO3* carries 2–4 peaks in every deep NR4A3 experiment and clears its panel in exactly one — the NORMAL parotid gland** ($p = 0.035$), not any carcinoma.** A signal present in normal tissue and absent from the tumours is the opposite shape from a tumour-driven one, and two nominal hits in 36 tests is what chance gives.

Two things this still does **not** say. NR4A3 in acinic cell carcinoma is **not the fusion and not EMC**: it is wild-type protein in a salivary-gland tumour, and NR4A1 — which supplies 8 of the 12 informative experiments — is a paralogue whose peak sharing with NR4A3 is 0.347 in matched dendritic cells. And the twelve ChIP-Atlas NR4A3 peak sets still say nothing at all: at 53–154 peaks they recover **no** panel gene, so their silence remains an absent reading. What the table establishes is that the surrogates cannot stand in for the missing experiment — now including a genuinely deep NR4A3 cistrome in the wrong disease — which is why §4.3's discriminating experiment remains **occupancy of an NR4A3 fusion in EMC chromatin** and not a re-analysis.

3.12 • What the instruments say together

Figure 4 puts the ordering on one screen. **ENO3 is supported by every instrument that returned a reading.** Both array platforms support it under an exact permutation test and after multiple-testing correction, as does every comparator stratum separately, including the myxoid-matched and reference-pool-matched arms. Matrix adjustment retains 75% of its delta on the platform where that covariate differs and 100% on the platform where it does not. It is in the top 2% of 14,120 genes in an independent cohort on an unrelated technology, the muscle markers are flat and more muscle-restricted than it is, and it carries more exact NBREs than its own composition-matched null. **The exception is the occupancy axis, and it is an exception for all three genes**: no class-A gene exceeds a background panel in any NR4A peak set (§3.11). **ENO3** holds the only two nominally significant values, and the one from an NR4A3 experiment falls in a **normal parotid gland** rather than any tumour — 2 hits in 36 tests against 1.8 expected, binomial p 0.54. **SEMA3C is the mirror image** — it fails the permutation test on both platforms, reverses sign with comparator choice, is $p = 0.84$ against pool-matched comparators, and carries no exact NBRE. **PPARG sits between them, and lower than it first appeared**, because its strongest cell is circular.

None of this converts association into causation for any of the three. Every axis here is correlative; the discriminating experiment (§4.3) remains unperformed; and ordering three genes by independent support is not evidence that any of them is bound by the fusion in EMC.

3.13 • No fourth EMC expression cohort — a second bounded negative

The most direct way to strengthen an n of 4, 6 and 10 is a fourth cohort. Six queries (§2.7) returned 56 GEO records, of which **22 were series or curated datasets** and the remainder individual sample and platform records. Every one of the 22 was read at sample level; none is a fourth EMC cohort.

Table 10. The cohort search, grouped by why each deposit is not a fourth cohort.

Table 10			
group	deposits	EMC samples	disposition
The three cohorts analysed here	GSE24369 (42 samples), GSE4303 (36), GSE28866 (99)	6, 10, 4	already used — and the search's positive control
The same EWS/NOR1 construct experiment	GSE11185 (4), GDS3481 (its curated view)	2 sample labels	HEK293 cells carrying a tet-inducible construct — not a tumour cohort
Other sarcoma and chondrosarcoma series	17 deposits, 4–51 samples each: GSE12475, GSE12592, GSE14469, GSE29085, GSE43045, GSE43632, GSE44934, GSE52677, GSE52679, GSE62747, GSE80126, GSE150474, GSE168560, GSE196000, GSE196002, GSE289237, GSE315379	0 in every one	read at sample level; no EMC sample in any

The first row is the result, not the preamble. It is the positive control: the same queries recovered all three cohorts already in use, and recovered their EMC arm sizes — 6, 10 and 4 — from GEO sample titles alone, an independent path to the three numbers Table 2 takes from the series matrices. A search that had failed to find them would have made the negative meaningless, so the negative is reported only because the control passed.

The seventeen zeros are what makes it a negative rather than an absence. They span chondrosarcoma profiling, myxoid liposarcoma, myxoinflammatory fibroblastic sarcoma, synovial-sarcoma-like tumours, clear cell sarcoma, Ewing sarcoma, rhabdomyosarcoma, translocation-sarcoma panels and two fusion-detection methods series — the adjacent territory in which an EMC sample could plausibly sit under a title that never names the disease. Two of them (GSE43632, *Large scale screening for fusion genes in sarcoma patient samples*; GSE80126) name no EMC token in title or summary yet were returned by the full-disease-name query, so GEO's [All Fields] index reaches text beyond the series prose — precisely the case in which a title-level screen would have recorded a false absence. Read at sample level, none of the seventeen carries an EMC sample.

What this bounds, and what it does not. The bound is reach, not existence: a deposit that names the disease nowhere in its GEO record is invisible to any term search, and a term search is not a systematic review — it does not reach other archives, controlled-access repositories, or supplementary tables of papers that never deposited at all. Within that reach, **no fourth EMC expression cohort exists** on GEO, and the three cohorts analysed here are the available public EMC *whole-transcriptome* record. So $n = 4, 6$ and 10 is a ceiling imposed by the disease's rarity rather than by the search (§5, Limitation 1).

— AND THE BOUND IS LOAD-BEARING, BECAUSE A DEPOSIT OUTSIDE IT EXISTS. This search asked GEO. It does not reach a study registered in the Sequence Read Archive that was never given a GEO series, and **one such study is public**. It is [PRJNA1357027 / SRP640302: 12 FFPE EMC tumour BioSamples](#), released 2025-11-11, all 12 runs downloadable, with **per-sample EWSR1 break-apart FISH status** (8 positive, 4 negative), site, size and morphology — larger than any cohort read here, and carrying the per-sample fusion annotation none of the three has. It has no linked publication and no GEO mirror, which is exactly why a GEO term search cannot see it. Characterisation, every query, the three transport controls and the raw payloads: [emc-fourth-cohort-sra-2026-08-08.md](#), artifact [emc-sra-study.json](#).

△ It is not a drop-in fourth arm, and Limitation 1 is unchanged for the analyses this paper runs. The deposit's `library_strategy` field says RNA-Seq while its own experiment title says **Targeted RNA-seq (TempO-Seq)** — a targeted panel, whose gene space is the panel's rather than the transcriptome's, and **the panel is named nowhere in the metadata**. A gene-set contrast run against it without naming the panel would return a matrix that is mostly zeros by construction, with nothing to warn the analyst. Reading EWSR1-negative as "not EMC" would be the same error in the other direction: a break-apart call names no partner, and a `TAF15::NR4A3` or `TCF12::NR4A3` case is EWSR1-negative by construction, so those four samples are **informative and unresolved**. What changes here is narrow and

it is a claim about this paper's own sentence: "*no fourth EMC cohort exists*" is no longer sayable without the GEO-side qualifier.

△ **One instrument fault is recorded rather than smoothed over, because it nearly halved this search.** Four of the six queries first returned exactly zero, and all four shared a field-restriction clause the two productive queries lacked — including one asking GEO for human chondrosarcoma expression series, a question it cannot honestly answer with nothing. Re-asked with the restriction lifted and the search terms unchanged, three of the four returned records (2, 4 and 32), taking the search from 7 series to 22; the fourth returned zero again and is the only zero here read as an absence. The negative above is the repaired search. The unrepaired one would have rested on two queries while reporting six (SI §S7).

!Figure 1

Figure 1. A set score means nothing until an arbitrary set of the same size is scored too. Grey histogram: 4,000 random gene sets of exactly the observed size, drawn from the platform's own mapped symbols under a fixed seed and scored identically to the real set. Shaded band: the central 95%. Vertical line: the observed delta. The annotation gives the value the set had to reach to clear the band and how far it got. Top row: the A+B direct-target set reaches 39% and 88% of its threshold. Bottom row: the published EMC phenotype overshoots by 11.9× and 4.2×. **This null controls the platform offset and set size, not gene–gene correlation**, so it is anti-conservative for coherent sets; §3.5 supplies the complementary exact label-permutation test.

!Figure 2

Figure 2. The entire published direct-target catalogue of an NR4A3 chimera is three genes. Counted across 2,276 retrieved full-text documents in five corpora (§3.11). **This is a count of what has been published and retrieved, not of what exists** — a claim about a search. Class B requires the transfer assumption that §3.2 shows failing in both directions.

!Figure 3

Figure 3. Every tumour, per gene and per comparator stratum. Each point is one tumour; the horizontal bar is the arm mean. Values are within-array z against that sample's own probe distribution. n = 6 EMC vs 29 comparators (GPL6244) and 10 vs 6 (GPL3290). **The two platforms measure different quantities — single-channel intensity and two-colour log-ratio against a reference pool — and are never pooled**, so no comparison across the two panels is licensed. The comparator strata are drawn separately because *SEMA3C*'s contrast changes sign between them (§3.6). No panel asserts that the fusion binds or drives any gene.

!Figure 4

Figure 4. Independent instruments applied to the three published direct targets. The columns are deliberately not commensurable and no glyph is scaled by effect size, so no area comparison across columns is possible: colour encodes only whether that instrument supported the gene, and each cell prints its own statistic in its own units. The amber cell marks the reading that is *circular* — *PPARG* on GPL3290, scored on the cohort from which high *PPARG* in EMC was first published (§3.8) — which is neither support nor absence. The 3SEQ column carries no test: at $n = 4$ it is a percentile within that deposit's own distribution. **The occupancy column is grey for all three genes**, and is the only column on which no gene is supported: it reports the best empirical p any of twelve informative NR4A experiments gives against a 198-gene background panel, judged at a Bonferroni threshold for those twelve. Eight are **NR4A1, a paralogue**; four are **wild-type NR4A3 in acinic cell carcinoma**, a different disease. Neither is the fusion (§3.11). **No cell asserts that the fusion binds or drives any gene**, and §3.11 records that no NR4A3-fusion cistrome has been retrieved and that nothing at all has been deposited on EMC material under any chromatin library strategy.

!Figure 5

Figure 5. The *ENO3* muscle-admixture objection, and its answer. *ENO3* is muscle-specific β -enolase and EMC arises in deep soft tissue of the limb. Horizontal axis: how muscle-restricted a gene is, as its mean within-array percentile in the two pooled skeletal-muscle RNA samples GSE24369 contains. Vertical axis: the EMC – comparator difference in within-array percentile points. **The two muscle samples are in neither arm and no contrast in this paper uses them**; they fix the scale only. Three markers more muscle-restricted than *ENO3* sit at or below zero. **This bounds admixture of differentiated skeletal muscle; it does not exclude a myogenic differentiation programme in the tumour itself**, which would move a marker with no contaminating tissue present.

4 • Discussion

4.1 • What these data are consistent with, and what they are equally consistent with

A target gene that is up in EMC is consistent with the fusion driving it, and equally consistent with: (a) EMC's cell of origin expressing it; (b) EMC's myxoid, hypocellular architecture; (c) a platform-wide offset; (d) the gene being a generic proliferation or matrix gene; (e) the anatomical site EMC arises in.

This version narrows more of that list than its predecessor did. The null calibration removes (c) and part of (d). **(b) is now partly measured rather than conceded.** The GPL6244 comparator arm is 23/29 myxoid, so it is largely matched to EMC on matrix architecture, and *ENO3* is unchanged against the myxoid-only arm (+0.808, $p = 8 \times 10^{-5}$). Adjusting for an 11-gene matrix proxy chosen to contain no EMC-selected gene leaves 75% of its delta where the covariate differs between arms and 100% where it does not. **(e) is bounded for *ENO3*** by the muscle control of §3.5. What remains genuinely unremoved is **(a)**: nothing in these datasets separates a gene the fusion drives from a gene EMC's cell of origin expresses, and the 3SEQ normal-organ arm does not help, because six visceral organs are not the soft tissue EMC arises in.

4.2 • What is new here

Four things, in descending order of what survives this paper's own worked example being wrong. Nothing here is a first-in-field claim.

- **The calibration, which is the contribution.** A size-matched empirical null on the platform's own genes converts a pervasive and uninformative "higher in the index arm" into a statement that can be refused — and it refuses this work's own aggregate, at a quantified distance (39% and 88% of threshold) rather than as a bare negative. It costs one seeded resampling. ****It is not specific to EMC, to this gene set, or to these three cohorts****; any series with a small index arm and a heterogeneous comparator arm has the same failure mode, and in rare tumours such series are the only ones that exist. Every biological claim below could be overturned tomorrow by a cistrome without touching this.
- **The map of what is missing, and the experiment that closes it.** Class A is three genes wide. Nothing has been deposited on EMC material under any chromatin library strategy, so no experiment has measured where an NR4A3 fusion binds or what chromatin does in EMC chromatin. The same archives, meanwhile, hold chromatin maps for EWSR1::WT1, EWSR1::ATF1, EWSR1::FLI1, FUS::DDIT3 and HEY1::NCOA2, and one *accessibility* screen carrying four NR4A3 fusions in HEK293T (GSE243553). And the 110 NR4A peak sets that do exist are measured — not assumed — to be unable to substitute (§3.11). §4.3 then names the discriminating experiment rather than gesturing at one. For a disease with no fusion cistrome, a specified missing experiment is a more useful output than another correlative reading, and it is the part of this paper addressed to anyone with a laboratory.
- **The confound audit.** Comparator composition read from the GEO sample titles rather than from a grouping label; the contrast recomputed against every stratum, against the reference-pool-matched comparators, and against a provenance-filtered matrix covariate; and a skeletal-muscle control for the one gene where that objection is obvious.
- **The ordering, which is the worked example and the weakest part.** The three genes are not equally supported: *SEMA3C* is supported by nothing that survives its own comparator being varied, and *PPARG*'s strongest reading is circular. $\triangle$ ****The surviving gene is the pre-designated positive control**** (Limitation 17), so *ENO3*'s elevation is not an independent finding of this work, and the ordering rests on three cohorts of 4, 6 and 10. Read it as a demonstration that the instrument discriminates where a raw contrast does not — not as a settled biology.

A note on what is *not* claimed. No prior source states the class-A count of three, but the reason is measured rather than asserted (§1.3), and it is not that the field says something different: the three genes are named in 3, 1 and 0 of 261 EMC review records, so there is no competing account to correct. $\triangle$ **That is a claim about a near-absence, the weaker of the two directions a citation index supports.** Seldom-named is a positive reading; *no source anywhere assembles them* is not established by it, and a review outside the searched corpus could do so.

4.3 • What would discriminate, named rather than hand-waved

1. **A cistrome in the right cell.** An NR4A3 ChIP-seq peak set with the *fusion* expressed, intersected with these expression reads: a gene that is up in EMC *and* carries a fusion-bound NBRE is driven; a gene that is up with no peak is correlated. The nearest existing dataset is Haller *et al.* (2019, *Nat Commun* 10:368, PMID 30664630): NR4A3 ChIP-seq in three human acinic cell carcinoma tumours, with a de-novo NBRE motif recovered in all three (processed data on Zenodo, doi 10.5281/zenodo.1483691; raw data on EGA, EGAS00001002795, controlled access). The caveat is load-bearing: acinic cell carcinoma carries *native* NR4A3 up-regulated by enhancer hijacking, not a fusion. Given §3.2's measurement that native NR4A3 does not activate the *PPARG* promoter the fusion does, that dataset answers "where does the NR4A3 DNA-binding domain go in a human tumour" and not "where does EWSR1::NR4A3 go". It must never be cited as the latter. The nearest dataset in which an NR4A3 *fusion* is the perturbation is GSE243553 (§3.11), and it is subject to the same restriction from the other direction: it reads chromatin **accessibility** in HEK293T, not occupancy in EMC chromatin, so it cannot be intersected with these expression reads to call a gene driven. The experiment named here is occupancy of an

NR4A3 fusion in EMC material, and §3.11 shows the field performs exactly that experiment routinely for neighbouring fusions.

2. **Fusion knockdown or degradation in a genuinely fusion-positive EMC model, with RNA-seq.** No such experiment was retrieved.
3. **Fusion-type-stratified EMC expression data.** Brenca *et al.* show class-3 versus class-4–6 semaphorins separating EWSR1- from TAF15-translocated EMC, but no readable series records which fusion each EMC sample carries, so every EMC arm here is a mixture and any fusion-specific signal is attenuated by an unknown amount.
4. **A within-EMC test against fusion level was attempted and does not discriminate at this n.** Holding disease constant and correlating each gene against NR4A3 level inside the EMC arm is the only axis in these data that speaks to fusion *output* rather than EMC membership. It gives $r = +0.37$ ($n = 6$) and -0.35 ($n = 9$) for ENO3 on the two platforms — opposite signs, no information. The GPL3290 axis reads 9 of that arm's 10 tumours, because it is anchored on NR4A3 and one sample carries no readable NR4A3 value. Reported so that the axis is not proposed again as though untried; NR4A3 array signal is in any case the 3' partner under a foreign promoter, not the fusion transcript.
5. **An NBRE motif scan** — performed (§3.10). It cannot demonstrate binding and did not resolve the question. **What remains undone on this axis is not another scan:** sequence cannot settle occupancy, and the discriminating experiment is item 1.

5 • Limitations

These are ceilings, not caveats: each one bounds what any sentence in §3 may be read to mean.

1. **n = 4, 6 and 10 EMC.** Nothing here survives being described as a distribution, and no result should be read as a population estimate. This is a ceiling on the disease, not on the search. A term search of GEO returned 56 records, of which 22 were series or curated datasets; every one was read at sample level, and none was a fourth EMC expression cohort. The seventeen unrelated sarcoma and chondrosarcoma deposits among them carry no EMC sample between them (§3.13, Table 10). The bound is what a term search can reach: a deposit naming the disease nowhere in its GEO record is invisible to it, and no other archive was searched.
2. **The three cohorts are never pooled, and must never be.** 3SEQ 3'-end read density is not array intensity; single-channel intensity and two-colour log-ratio are not the same quantity either. The concordance in §3.5–3.7 is sign agreement across three independent measurements, which is weaker than a combined estimate and is deliberately reported as the weaker thing.
3. **Transcript, not protein.** *SGK1* is the worked example: its published protein and transcript directions oppose, and this instrument can only see the second.
4. **No occupancy, and therefore no causality from the fusion.** Nothing here shows any gene being bound by EWSR1::NR4A3 in EMC. The class-A assays were performed in engineered human fibroblasts, in rat chondrogenic cells, and with a chimera (TFG::NR4A3) that is not the common one.
5. **The normal arm is a six-organ tissue panel, not matched adjacent tissue,** so it cannot separate EMC-specific from mesenchymal-lineage-specific.
6. **Comparator arms differ between platforms.** §3.6 turns this into a designed contrast rather than only a limitation, but the strata are small — 6 desmoids, 3 DFSP, 3 GIST — and a stratified contrast at $n = 3$ comparators can report no p below 0.0035 however large the effect.

7. **Fusion type is unrecorded in every series**, so each EMC arm mixes EWSR1::NR4A3 with whatever TAF15::NR4A3 and rarer variants it contains, attenuating any fusion-specific signal by an unknown amount.
 8. **Multiple testing is corrected for the per-gene permutation results only**. The size-matched empirical p-values and the set-level and stratified permutation p-values remain uncorrected; the stratified panel in particular reports five contrasts per gene on GPL6244 (three comparator classes plus their myxoid and non-myxoid aggregations, which are not independent of the classes), which is why §3.6 reads it on the least favourable stratum rather than the best.
 9. **The size-matched null is competitive and anti-conservative for coherent sets**; the exact permutation null is self-contained but cannot make three genes into more than three genes. The two disagree for *SEMA3C*, which is reported rather than reconciled.
 10. **GPL3290 is relative**, and its comparator arm spans two reference pools (§3.4). Only the between-group contrast is interpretable, never an absolute level, and its probe → symbol mapping runs through an EST accession bridge, so a gene unreadable there may be absent from the bridge rather than from the array.
 11. **The 3SEQ rows are medians over very few peaks** — 2 (*ENO3*), 3 (*SEMA3C*), 5 (*PPARG*) — and the percentile calibration is a rank within one deposit, not a test.
 12. **The covariate adjustment is a sensitivity analysis, not a correction**. If a panel gene is itself driven by the fusion the adjustment removes real signal; that possibility is not excluded, only made unlikely by the provenance filter.
 13. **The muscle control bounds differentiated-muscle admixture only**, not a myogenic programme intrinsic to the tumour.
 14. **The motif scan speaks to sequence, never to occupancy**, is restricted to a fixed −10 kb/+15 kb window, and does not reproduce the published site coordinates for either *ENO3* or *PPARG*.
 15. **Nothing here is an efficacy, selectivity, safety, therapeutic-window or clinical-readiness claim** for any agent, target or gene, and expression data cannot become that evidence. No drug, dose, schedule or patient population is named or implied.
 16. **The occupancy axis is measured on the wrong protein or the wrong disease, in every experiment**. §3.11 is not evidence that these genes are unbound by the fusion. Eight of the twelve informative experiments are NR4A1, a paralogue sharing 0.347 of its peaks with NR4A3 in matched cells; the other four are **wild-type NR4A3 in acinic cell carcinoma**, where the protein is activated by enhancer hijacking rather than fused, in a salivary-gland lineage that is not EMC. *PPARG*'s zero in all four is a real negative about that setting and says nothing directly about EMC — §3.2 records native NR4A3 failing to activate the very promoter the fusion activates, so a native cistrome is expected to disagree with a fusion one at exactly this gene. The axis bounds what surrogates can show and nothing more.
 17. **The positive control and the surviving result are the same gene**. *ENO3* was designated the instrument's positive control before any biological read (§2.4) and is the gene that survives every subsequent test (§3.12). Its elevation in EMC is therefore not an independent finding of this work — it was selected as the control because it was expected to be elevated. §3.5 sets out why the rest of the reading is not thereby circular: the control tested one proposition, and none of the tests that separate the three genes was part of it. The structural weakness remains, and the observation that would remove it is a gene other than *ENO3* clearing the same bar.
-

6 • Conclusion

A gene-set read on a small rare-tumour series is uninterpretable until an arbitrary set of the same size has been scored beside it on the same platform. That is the general claim, it costs one seeded resampling, and this paper's worked example shows what it buys: the aggregate direct-target set reaches 39% and 88% of its size-matched threshold and does not clear, while the published EMC transcriptional phenotype clears the same threshold 11.9-fold and 4.2-fold on the same instrument in the same run. The instrument demonstrably reads the disease and does not read the set — a discrimination no raw contrast on these data makes, because on GPL3290 a raw contrast calls almost everything significant.

Applied to the three genes, it separates what is otherwise treated alike. *ENO3* is elevated on both readable array platforms under an exact permutation test and after multiple-testing correction, against every comparator stratum separately including the myxoid-matched and reference-pool-matched arms, in the top 2% of 14,120 genes in an independent cohort on an unrelated technology, with a skeletal-muscle admixture control that does not explain it. *PPARG*'s strongest reading is circular and what remains does not survive correction; *SEMA3C* survives none of these tests and changes sign with the choice of comparator. Δ *ENO3* was also the pre-designated positive control, so its elevation is not an independent finding of this work (Limitation 17), and the ordering rests on cohorts of 4, 6 and 10. It is a demonstration of the instrument, not a settled result.

The binding constraint on the biology is not sample size and not statistics. It is that class A is three genes wide, and that **no experiment has measured where an NR4A3 fusion binds, or what chromatin does, in EMC material** (§3.11 — a bounded statement about what has been deposited under a label an archive indexes, not a claim that no such data exists anywhere). The one genome-wide chromatin readout that carries NR4A3 fusions at all reads *accessibility* in HEK293T (GSE243553), not occupancy in EMC chromatin, and cannot close the gap. Nor can the existing NR4A chromatin data stand in for it: across 110 peak sets — including four deep NR4A3 cistromes in acinic cell carcinoma, a disease driven by wild-type NR4A3 — no class-A gene carries occupancy beyond a background panel. ★ **The sharpest form of that negative is comparative: the field performs this experiment routinely for the sibling fusions — EWSR1::WT1, EWSR1::ATF1, EWSR1::FLI1, FUS::DDIT3 and, twice, HEY1::NCOA2 — and has never performed it on EMC material.** Until it is, "up in EMC" and "driven by the fusion" cannot be told apart for any gene named here, *ENO3* included. **That experiment is specified in §4.3 and is the one thing that would change any of this.** No further correlative re-analysis of these deposits will.

Data and code availability

All primary data are public and no new data were generated. Every analysis is CPU-only and reproduces without a login, a rental or a GPU.

Public datasets.

accession	platform	source	primary publication
GSE24369	GPL6244	NCBI GEO	linked series PubMed identifier 21536545
GSE4303	GPL3290	NCBI GEO	Subramanian <i>et al.</i> , <i>J Pathol</i> 2005;206:433–444 (PMID 15920699)
GSE28866	3SEQ / GPL10999	NCBI GEO series supplementary peak tables	Brunner <i>et al.</i> , <i>Genome Biol</i> 2012;13(8):R75 (PMID 22929540)

Gene-set libraries are served through Enrichr (Kuleshov *et al.*, 2016): ChEA_2022, TRRUST_Transcription_Factors_2019, TF_Perturbations_Followed_by_Expression and MSigDB_Hallmark_2020. Each term used is pinned verbatim in the code. Referenced but not used as data: Haller *et al.* (2019) NR4A3 ChIP-seq — processed

data Zenodo doi 10.5281/zenodo.1483691 (open), raw data EGA EGAS00001002795 (controlled access); §4.3 states why it does not answer this question. The NR4A ChIP-seq census of §3.11 reads public ChIP-Atlas and ReMap2022 records, including GSE186199.

Code and derived artifacts. All are openly available in the project repository (<https://github.com/trimcrae/rare-cancers>), which will be archived to Zenodo with a citable DOI at submission:

artifact	producer
nr4a3-fusion-targets.json — evidence table, global offsets, null calibrations, per-gene and per-set scores, controls, circularity grade	nr4a3_fusion_targets.py
emc-expression-panels.json → gene_reads — the independent second implementation of the per-gene array reads	emc_expression_panels.py
gse28866-tumour-vs-normal.json → per_gene.values and ratio_calibration — the 3SEQ arm and its percentile calibration against all 14,120 genes in the deposit	gse28866_tumour_vs_normal.py
nr4a3-fusion-targets-robustness.json — exact label-permutation p-values, leave-one-out jackknife, rank-based re-read and BH q-values	nr4a3_fusion_targets_robustness.py
nr4a3-fusion-targets-confounds.json — comparator composition, the muscle-admixture control, every stratified and reference-pool-matched contrast with its own exact permutation p, the covariate-adjusted sensitivity analysis, minimum detectable effects, and the within-EMC axis	nr4a3_fusion_targets_confounds.py
nr4a3-fusion-targets-occupancy.json — NR4A ChIP-seq occupancy at the class-A genes across 110 peak sets, each count calibrated against a 198-gene background panel, with the depth rule that marks an undetectable peak set uninformative rather than negative and the antigen rule that scores only NR4A ChIPs	nr4a3_fusion_targets_occupancy.py (reads the committed emc-ret-cistrome.json and its cached peaks; no network)
figures/fig1–fig5 (PNG + PDF) and figures/figure-provenance.json	nr4a3_fusion_targets_figures.py
emc-ret-target-scan.json → part_1_nbre_scan — NBRE/NurRE counts, the dinucleotide-preserving shuffle null and the background-panel ranks. Ensembl sequences are cached, so the scan re-derives offline	emc_ret_target_scan.py
offline arithmetic guards	tests/test_nr4a3_fusion_targets.py, tests/test_nr4a3_fusion_targets_confounds.py, tests/test_nr4a3_fusion_targets_figures.py

The null draw is seeded (20260807) and the pool size, seed and universe are recorded per platform, so every empirical p is reproducible from the committed code and the public accessions alone. Each producer re-derives its inputs and refuses to write if any row disagrees with the artifact that owns it; the figure generator stamps the content hash of every artifact it read.

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Conflicts of interest

The author declares no competing interests.

Ethics approval and consent

Not required. This study analysed only publicly available, de-identified gene-expression deposits and generated no new human or animal data. No individual is identifiable from anything reported here.

Author contributions

T.D.M. is the sole author and is responsible for the study design, analysis, interpretation and manuscript.

Use of generative AI

This work was carried out with the assistance of a large language model-based research agent (Claude, Anthropic), used for literature extraction, analysis code, statistical computation and manuscript drafting. All analyses are executed by committed, deterministic, independently re-implemented and offline-reproducible code (see Data and code availability), and were reviewed by the author, who takes full responsibility for the content. No AI tool is listed as an author, in accordance with journal policy.

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None.

Appendix A · Superseded values and corrected claims

Retained so that a superseded number stays quotable as history and not as a current fact.

claim, as previously written	status	what replaced it
"That pattern is the shape of a platform-wide offset."	superseded 2026-08-07	The global offset is -0.0084 SD (GPL6244) and $+0.0258$ SD (GPL3290), an order of magnitude below the effects in question. The remedy is unchanged — the null absorbs both — but the mechanism is null-band <i>width</i> at these arm sizes, not offset (§3.4).
GSE24369's comparator arm described as containing "6 fibrosarcoma", and the comparators as "dense".	corrected 2026-08-08	The GEO titles are Myxofibrosarcoma 1–6; 23 of 29 comparators are myxoid (§2.2). The earlier wording carried an internal grouping label into a dataset description and inverted the premise of confound (b).
"PPARG ... significant on one platform" reported as independent support.	corrected 2026-08-08	PPARG on GPL3290 is circular : GSE4303 is the cohort from which high PPARG in EMC was published (§3.8).
Every $p_{\text{emp}} = 0.0005$ written as an equality.	corrected 2026-08-08	0.0005 is the resolution floor of a 4,000-draw two-sided null (2/4001) and is written ≤ 0.0005 (§2.3).
"Four of four graded controls agree", with PLAGL1/GPL6244 marked "not graded".	corrected 2026-08-08	Five of six control $\times$ platform cells are computable and all five agree; PLAGL1/GPL6244 is <i>inside its null band</i> and is not a reading at this power. The three-state grading rule is now stated in §2.4.
"Deep NR4A1 sets (ReMap2022) do recover both SEMA3C and ENO3" — offered in §3.11 as a near-miss worth noting.	corrected 2026-08-08	True and uninformative: 82.8% of a 198-gene background panel is also recovered by that catalogue. Calibrated, no class-A gene exceeds the panel in any NR4A peak set (§3.11, Table 9). The uncalibrated version was the same error §1.1 exists to refuse, inside the section that reports the paper's central negative.
A background citation attributing the cloning of the EMC fusion to a 1995 paper.	withdrawn, then re-anchored 2026-08-08	The original PMID traced to no held source and was written from recollection, and the statement was re-anchored on the GEO series record and Brenca <i>et al.</i> $\triangle$ That left §1.3's "the hypothesis is thirty years old" resting on a date with no source in the record — a dated claim escapes an identifier checker because a bare year carries no identifier. The cloning paper is now retrieved rather than recalled (PMID 8634690, 1995, reference 9a).
The title "The direct-target catalogue of EWSR1::NR4A3 is three genes wide, and one gene survives calibration", and a §1 that reached the calibration third.	superseded 2026-08-08	The paper led with its weakest result. ENO3 is the pre-designated positive control (Limitation 17), the ordering rests on cohorts of 4, 6 and 10, and no gene here is separable from disease association at all — while the size-matched null is general, reusable beyond this disease, and unaffected by any of that. Retitled around the calibration; §1 now opens on the failure mode, the gene ordering is stated as the worked example in §4.2 and §6, and no number in §3 changed.
"Two questions are routinely conflated " (§1.3) and "the field's prose does not usually say so " (§4.2).	superseded 2026-08-08	Both asserted what a literature does, and neither had been measured. Measured: the three class-A genes are named in 3, 1 and 0 of 261 EMC review records. A literature that seldom names these genes cannot routinely conflate claims about them; the true state is a near-absence of any account, not a mistaken one, and §1.3 and §4.2 now say that instead. The scientific results are untouched — nothing in §3 ever depended on what the field believed.
"...never assembled and tested against a proper calibration" (cover letter).	corrected 2026-08-08	An unrestricted negative about all prior literature, resting on nothing. Narrowed to what was actually done — a search, reported as a search — and the novelty claim now rests on the measured near-absence rather than on an assertion that no prior assembly exists.
"All twelve NR4A3-specific peak sets are too shallow to recover any gene at all", and the occupancy axis reported across 86 peak sets, 8 informative experiments and 24 tests.	superseded 2026-08-08	True of the twelve ChIP-Atlas sets and still stated of them, but no longer true of the axis: the Haller <i>et al.</i> acinic cell carcinoma deposit adds four NR4A3 cistromes at 8,501–18,666 peaks, 55–121 $\times$ the deepest previously available. The axis is now 110 peak sets, 12 informative experiments and 36 tests (§3.11, Table 9). The conclusion is unchanged and better supported — 2 hits against 1.8 expected, binomial p 0.54 — and PPARG's zero is now a negative rather than an absent reading.
The occupancy verdict decided by comparing the observed hit count with its expected value.	corrected 2026-08-08	Expected is fractional and observed is an integer, so 2 against 1.8 read as an excess. The count is judged by a binomial tail ($p = 0.54$ for this many or more); no wording in any version of this paper rested on the earlier comparison.
"No genome-wide chromatin experiment performed with an NR4A3 fusion was retrieved in 2,276 full-text documents across five corpora" — §3.11, §4.2, §6, the abstract ("no chromatin experiment with an NR4A3 fusion was retrieved") and the cover letter, each reading the corpus count as an absence.	retracted 2026-08-08	The corpus count is unchanged and was never wrong : 2,276 documents were searched, 153 name both a genome-wide chromatin method and NR4A3/NOR-1/TEC, and none of the 153 applies one to an NR4A3 chimera. What is retracted is the inference from that screen to an absence . A wider search the same day — the primary sequence archives rather than the literature alone — retrieved GEO GSE243553 (PMID 39048711, public 2024-07-24), a pooled single-cell ATAC screen in HEK293T whose library carries EWSR1-NR4A3, TAF15-NR4A3, TCF12-NR4A3 and TFG-NR4A3 with wild-type NR4A3 and the reciprocal NR4A3-EWSR1 as controls. The earlier screen could not reach it for two reasons, both recorded: the paper's title and abstract say only ">100 oncofusions" and name NR4A3 nowhere, and this project's prior chromatin census (emc-ret-cistrome.json) filtered on antigen $\in \{\text{NR4A1, NR4A2, NR4A3}\}$ with a ChIP-seq-only method vocabulary, which no pooled ATAC deposit can satisfy. The replacement claim is narrower and is what §3.11, §4.2, §6 and the abstract now carry : <i>no experiment has measured where an NR4A3 fusion binds, or what chromatin does, in EMC material</i> — GSE243553 being accessibility rather than occupancy, and HEK293T rather than EMC — while the same archives hold chromatin maps for EWSR1::WT1, EWSR1::ATF1, EWSR1::FLI1, FUS::DDIT3 and (twice) HEY1::NCOA2. Full search record: nr4a3-cistrome-search-2026-08-08.md ; corpus: lit-targets-nr4a3-cistrome.json . No number in §3 moved and the occupancy axis of Table 9 is untouched.

claim, as previously written	status	what replaced it
§3.10's first caveat: " A motif is not occupancy — only a chromatin experiment shows binding, and §3.11 records that none exists for any NR4A3 fusion."	superseded 2026-08-08	The second clause inherited the retracted absence directly above and was false the moment it was retracted. The experiment exists, and it has now been <i>run against these exact coordinates</i> rather than merely cited: the four exact NBREs were intersected with GSE243553's per-fusion accessibility calls on the matching build (deposit hg38, coordinates GRCh38, no lift-over). Three of four sites fall inside TAF15-NR4A3's intervals and none inside EWSR1-NR4A3's, and neither observation supports the motif argument — the EWSR1-NR4A3 set recovers only 2 of 203 background promoters, so its zero is not a reading, and the TAF15-NR4A3 co-location does not clear a null that slides the whole four-site configuration at its true spacing ($p = 0.08$, 20,000 seeded draws). Δ The calibration is the finding, not the overlap. Two of the four sites are 153 bp apart and one 500 bp interval covers both, so nulls treating the sites as independent return $p \leq 0.002$ and overstate the result =40-fold; the geometry-preserving null is the honest one and it does not clear. Bounded three ways in the live text: HEK293T rather than EMC chromatin, accessibility rather than binding, and <i>ENO3</i> is this paper's own pre-designated positive control (§2.4), so a hit would have been a check on the instrument rather than a discovery. Method, seeds and all four nulls: gse243553-eno3-overlap-2026-08-08.md . No number in §3.10's motif counts changed.
§3.13: "Within that reach, no fourth EMC expression cohort exists ", and "the three cohorts analysed here are the available public EMC transcriptional record".	narrowed 2026-08-08	The search was GEO-side and its reach caveat already said so; what it did not say is that a deposit outside that reach had been found. PRJNA1357027 / SRP640302 is public in the Sequence Read Archive — 12 FFPE EMC tumour BioSamples , released 2025-11-11, all 12 runs downloadable, with per-sample EWSR1 break-apart FISH status (8+/4-) — with no linked publication and no GEO mirror, which is precisely why a GEO term search cannot see it. Δ Limitation 1's n = 4, 6 and 10 is UNCHANGED for the analyses this paper runs , because the deposit is TempO-Seq targeted-panel data whose panel is named nowhere in its metadata while its <code>library_strategy</code> field reads RNA-Seq — a whole-transcriptome contrast run against it would return a matrix mostly zeros by construction with nothing warning the analyst. Nor may EWSR1-negative be read as "not EMC": a break-apart call names no partner and a TAF15::NR4A3 case is EWSR1-negative by construction, so those four samples are informative and unresolved. What is retracted is only the unqualified sentence. Characterisation, every query and the three transport controls: emc-fourth-cohort-sra-2026-08-08.md .

Appendix B · What would change this paper's conclusions

observation	what it would overturn
An EWSR1::NR4A3 cistrome showing no peak near <i>ENO3</i>	The only remaining reading under which <i>ENO3</i> is a direct fusion target; it would move <i>ENO3</i> to "up in EMC, not fusion-bound".
An EWSR1::NR4A3 cistrome showing a peak near <i>SEMA3C</i>	Would restore <i>SEMA3C</i> as a direct target despite its failure on every correlative axis here, and would show that comparator-driven expression contrasts can mask a real target.
A fusion-positive EMC model with fusion knockdown and RNA-seq	Would replace every association in this paper with a directional test, and could overturn all three orderings at once.
An EMC expression series recording fusion type per sample	Would test whether <i>SEMA3C</i> 's comparator-dependence is really EWSR1-versus-TAF15 heterogeneity inside the EMC arm.
A soft-tissue normal comparator arm	Would remove confound (a), the one this paper cannot narrow.
Any deep NR4A3 ChIP-seq in any human tissue	Would make the §3.11 depth argument testable rather than a bounded negative about a search.
A per-arm reanalysis of GSE243553's barcode → variant files placing the four NR4A3-fusion arms' accessibility calls against a background panel	Would convert §3.11's quoted figures into measured ones and give the first calibrated chromatin read of an NR4A3 <i>fusion</i> — in HEK293T, so it would still not make any gene here fusion-driven in EMC.
Any chromatin experiment deposited on EMC material under a label an archive indexes	Would remove the bound §3.11 places on its own negative, which is reach and not existence.

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